

A Unified, Invariant, Geometry-Based Measurement Theory for Machine Cognition: NCAP-10

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Abstract

NCAP-10 is presented as a management-accessible measurement framework designed to evaluate how different AI and machine-cognitive systems operate, adapt, and behave, regardless of their underlying architecture. Unlike existing evaluation methods that depend on specific model designs, task benchmarks, or human-referenced scoring systems, NCAP-10 provides a common foundation for assessing systems that may function in fundamentally different ways, including neural networks, neuromorphic processors, analog accelerators, hybrid cognitive systems, and emerging computational substrates.

The framework treats cognition as a dynamic process rather than a collection of task outcomes. Accordingly, NCAP-10 evaluates systems by examining their behavior over time, including stability under load, efficiency of energy use, synchrony of internal processes, adaptability to new conditions, robustness of memory structures, organization of internal attractor states, and the capacity to regulate behavior when constraints change. These properties are interpreted as observable system behaviors rather than intelligence scores or human-like performance metrics.

NCAP-10 ensures that measurements remain comparable even when systems differ substantially in architecture, hardware, or internal representation. It achieves this by defining a set of observable indicators, structural embeddings, and invariant quantities that capture the essential signatures of a system's cognitive dynamics. These indicators allow meaningful comparison across diverse technologies and across successive versions of the same system.

The framework does not attempt to replicate psychometric testing or estimate human-style intelligence quotients. Instead, it identifies stable, architecture-independent markers of machine cognition, including measures of stability, synchrony, energy efficiency, attractor structure, plasticity, memory robustness, and adaptive behavior. These markers form a consistent basis for evaluating how a system processes information, maintains coherence, and responds to changing conditions.

By grounding evaluation in system dynamics rather than human-centered assumptions, NCAP-10 establishes a unified measurement framework from which benchmarking procedures may be developed. This enables organizations to compare different AI systems, track improvements across versions, assess the impact of hardware or training changes, and evaluate emerging machine-cognitive architectures using a consistent and theoretically grounded approach.

1 Technical Preface:

NCAP-10 is formulated as an architecture-invariant measurement theory for machine cognition. It introduces a manifold-valued cognitive object

$$N = (X, p(m \mid d), g_I, T),$$

where X lies on a differentiable manifold $\mathcal{M}$ equipped with a specified topology, coordinate charts, and an information-geometric metric g_I . From the admissible space of such objects, denoted $\mathcal{N}$, a family of measurable cognitive observables, difficulty embeddings, and invariant quantities is defined through operators with explicit codomains:

$$\Phi : \mathcal{N} \rightarrow R^k, \quad \Psi : \mathcal{N} \rightarrow \mathcal{D}, \quad \Omega : \mathcal{N} \rightarrow \mathcal{I}.$$

Architecture invariance is established through representation-preserving transformations. For any admissible map $F : A_1 \rightarrow A_2$ between system representations, the induced pullback satisfies a quasi-isometric compatibility condition,

$$F^* g_I \sim g_I, \quad p_1(m \mid d) = p_2(F(m) \mid F(d)),$$

ensuring that $N(A_1)$ and $N(A_2)$ remain equivalent up to allowable distortion. All observable quantities are required to commute with such transformations:

$$\Phi(N(A)) = \Phi(N(F(A))), \quad \Psi(N(A)) = \Psi(N(F(A))), \quad \Omega(N(A)) = \Omega(N(F(A))),$$

which guarantees invariance of the derived measurements.

The paper presents the present mathematical formulation of NCAP-10, its invariance axioms, and the associated observability operators. Standardized cross-architecture benchmarking emerges through comparison of invariant observable coordinates, providing practical applicability while remaining grounded in the theoretical framework.

2 Introduction

Existing AI evaluation frameworks typically rely on architecture-specific assumptions, task-defined benchmarks, or human-normed psychometric constructs.

While effective within restricted domains, such approaches are not generally designed to provide a unified basis for comparing heterogeneous computational systems—such as transformer models, neuromorphic processors, analog accelerators, hybrid cognitive systems, or emerging alternative computational substrates—within a single measurement structure.

NCAP-10 addresses this limitation by introducing a geometry-based, generative, and architecture-invariant measurement framework grounded in information geometry, causal representation principles, robustness analysis, and system dynamics. Rather than evaluating systems through discrete task outputs or human-referenced performance scales, NCAP-10 characterizes cognition as a measurable dynamical process defined on a geometric state space represented as a structured manifold.

The framework evaluates machine and neuromorphic systems through system-level dynamical properties, including stability under load, energy-flow efficiency, timing fidelity and synchrony, plasticity and adaptation, memory robustness, attractor organization, regulatory mechanisms, and reasoning under changing constraints. These quantities are treated as observable features of cognitive dynamics rather than behavioral outcomes or task scores.

Accordingly, NCAP-10 is not a psychometric instrument and does not compare human populations, developmental groups, or clinical cohorts. Traditional psychometric frameworks evaluate individuals relative to normative populations and produce comparative behavioral scores. In contrast, NCAP-10 evaluates computational systems relative to invariant dynamical criteria that are independent of human performance distributions.

This formulation enables comparison across multiple levels of system organization, including:

- **Implementation level:** computational architecture, hardware substrate, energy allocation.
- **Learning level:** training regime, adaptation mechanisms, regulatory adaptation.
- **Memory level:** memory organization, robustness, attractor structure.
- **Regulation level:** control mechanisms, stability properties, constraint handling.

The objective is not to estimate intelligence quotients or replicate human assessment protocols, but to quantify invariant signatures of machine cognition that persist across representational and architectural variation.

From these measurements, NCAP-10 derives system-level indices—including stability metrics, synchrony measures, energy-efficiency indicators, attractor dominance characteristics, plasticity profiles, memory robustness indicators, and adaptation scores—that arise as observable mappings under Φ , embedding operators under Ψ , and invariant quantities under Ω . These indices are interpreted as dynamical observables within the measurement framework rather than psychometric constructs.

By grounding evaluation in invariant system dynamics rather than human-centered assumptions, NCAP-10 establishes a unified measurement framework from which benchmarking procedures may be constructed for comparing heterogeneous cognitive architectures across implementation substrates while maintaining theoretical consistency, empirical interpretability, and cross-architecture comparability.

3 Primitive Object: The NCAP-10 Cognitive Object

NCAP-10 represents an intelligence system through a structured cognitive measurement object composed of four fundamental components. These components specify the system’s difficulty space, its probabilistic measurement behavior, its information-geometric sensitivity, and the transformations that preserve its equivalence-class membership under the NCAP-10 invariance axioms. Formally, the primitive object is written as

$$N = (X, p(m \mid d), g_I, T).$$

- **Difficulty Space X :** a subset $X \subseteq \mathcal{M}$ of a differentiable manifold $\mathcal{M}$, whose points represent task or environmental difficulty states under which the system may operate.
- **Measurement Behavior $p(m \mid d)$:** a conditional measurement distribution over observable outcomes m given a difficulty state $d \in X$. This distribution characterizes how the system responds under varying difficulty conditions.
- **Information Metric g_I :** an information-geometric metric, typically derived from Fisher information, that quantifies the local distinguishability of neighboring measurement distributions and the induced sensitivity of the system’s responses to perturbations.
- **Invariant Transformations T :** a set of admissible endomorphisms

$$T \subseteq \text{End}(N), \quad F(N) = N',$$

where $N \equiv_{\varepsilon} N'$ denotes equivalence under the NCAP-10 invariance axioms with allowable bounded distortion. These transformations modify the system’s representation while preserving its equivalence-class membership.

Together, these components define the geometric measurement structure associated with the manifold, forming the foundational object on which NCAP-10 evaluates machine cognition.

4 Formal Definitions (Management Version)

This section provides a non-technical description of the core components that make up the NCAP-10 measurement framework. The purpose is to explain, in clear terms, how NCAP-10 represents an intelligence system, how it interprets system behavior, and how it ensures that different architectures can be compared fairly.

Difficulty Space

NCAP-10 assumes that every system operates under a range of task or environmental conditions. These conditions form the *difficulty space*, which represents all situations the system may encounter. Each point in this space corresponds to a specific type or level of challenge.

Measurement Behavior

For every difficulty condition, a system produces observable outcomes. NCAP-10 models this through a *measurement behavior*, which describes how the system’s outputs vary as the difficulty changes. Instead of focusing on single outcomes, NCAP-10 considers the full range of possible responses and how likely each one is.

Information Geometry

NCAP-10 uses ideas from information geometry to quantify how sensitive a system is to changes in difficulty. This sensitivity is captured through an *information metric*, which measures how easily the system’s behavior can be distinguished when conditions shift. Systems that react strongly to small changes have high sensitivity; systems that remain stable have lower sensitivity.

Cognitive Object Space

Every system described by NCAP-10 is represented as a structured object containing its difficulty space, measurement behavior, information metric, and allowable transformations. The collection of all such objects forms the *NCAP-10 cognitive object space*. This space contains every system that satisfies the NCAP-10 requirements.

Admissible Transformations

Different systems may use different internal representations, architectures, or encodings. NCAP-10 accounts for this by defining *admissible transformations*: changes to a system’s representation that do not alter its essential behavior. These transformations allow NCAP-10 to compare systems even when they look very different internally.

NCAP-10 Equivalence

Two systems are considered *equivalent* under NCAP-10 if an admissible transformation can map one system’s representation to another while preserving its key behavioral and geometric properties. This equivalence ensures that NCAP-10 evaluates systems based on what they do, not how they are implemented.

Observable Operators

NCAP-10 defines three types of observable outputs:

- measurable indicators of system behavior,
- representations of difficulty or task structure,
- quantities that remain stable across different system implementations.

These observables are defined so that they depend only on the system’s equivalence class, meaning they remain unchanged under admissible transformations. This guarantees that NCAP-10’s measurements are fair, consistent, and architecture-independent.

5 Formal Definitions (Technical Version)

This section introduces the formal mathematical objects underlying NCAP-10. The goal is to specify the difficulty space, measurement behavior, information geometry, admissible transformations, the induced equivalence relation, and the observable operators that together constitute the NCAP-10 measurement framework.

[Difficulty Manifold] Let $\mathcal{M}$ be a differentiable manifold equipped with topology τ and atlas $\mathcal{A}$. The *difficulty space* is a subset $X \subseteq \mathcal{M}$ whose points represent task or environmental difficulty states under which a system may operate.

[Measurement Space] Let $\mathcal{O}$ denote the space of observable measurement outcomes. A *measurement behavior* is a conditional probability distribution

$$p(m \mid d) : \mathcal{O} \times X \rightarrow [0, 1],$$

assigning to each difficulty state $d \in X$ a distribution over observable outcomes $m \in \mathcal{O}$.

[Information Metric] Let $\{p(\cdot \mid d)\}_{d \in X}$ be a family of conditional measurement distributions. The *information metric* g_I is the Fisher information metric induced on X , defined by

$$g_I(d)_{ij} = E_{m \sim p(\cdot \mid d)} \left[\frac{\partial}{\partial d_i} \log p(m \mid d) \frac{\partial}{\partial d_j} \log p(m \mid d) \right],$$

which quantifies the local distinguishability of neighboring measurement distributions and the induced sensitivity of system responses.

[Cognitive Object Space] The *NCAP-10 cognitive object space* is the set

$$\mathcal{N} = \{ N = (X, p, g_I, T) \mid N \text{ satisfies the NCAP-10 axioms} \}.$$

[Admissible Transformations] An *admissible transformation* is an endomorphism

$$F : \mathcal{N} \rightarrow \mathcal{N}, \quad F(N) = N',$$

such that N' satisfies the NCAP-10 invariance axioms and remains within an allowable bounded distortion class relative to N . The set of all such transformations is denoted

$$T \subseteq \text{End}(\mathcal{N}).$$

[NCAP-10 Equivalence] NCAP-10 equivalence is the equivalence relation induced by the admissible transformation set T . Two cognitive objects $N, N' \in \mathcal{N}$ are *NCAP-10 equivalent*, written $N \equiv_\varepsilon N'$, if there exists $F \in T$ such that

$$F^* g_I \sim g_I \quad \text{and} \quad p(m \mid d) = p'(F(m) \mid F(d)),$$

where $\sim$ denotes quasi-isometric compatibility with bounded distortion ε .

[Observable Operators] Let $[N]$ denote the equivalence class of N under $\equiv_\varepsilon$. NCAP-10 defines three observable operators acting on equivalence classes:

$$\Phi : \mathcal{N} / \equiv_\varepsilon \rightarrow R^k, \quad \Psi : \mathcal{N} / \equiv_\varepsilon \rightarrow \mathcal{D}, \quad \Omega : \mathcal{N} / \equiv_\varepsilon \rightarrow \mathcal{I}.$$

Each operator is required to be invariant on equivalence classes:

$$\Phi([N]) = \Phi([F(N)]), \quad \Psi([N]) = \Psi([F(N)]), \quad \Omega([N]) = \Omega([F(N)]),$$

for all admissible transformations $F \in T$.

Together, these definitions specify the formal mathematical structure underlying NCAP-10 and provide the foundation for its invariance theorems and observable measurement framework.

Section Summary

Together, these components form the formal structure of NCAP-10. They provide a unified way to describe how a system behaves under different conditions, how sensitive it is to change, how its representation may vary, and how meaningful, comparable measurements can be extracted across diverse cognitive architectures.

6 Operational Implications

This section provides an operational interpretation of NCAP-10. All quantities described below are derived summaries of the formal geometric measurement structure and are defined on equivalence classes in $\mathcal{N} / \equiv_\varepsilon$.

Canonical Difficulty Structure

NCAP-10 represents difficulty as a structured state in a geometric space $X \subseteq \mathcal{M}$. Difficulty is not a scalar value or aggregate score, but a multi-dimensional condition that specifies the operating context of a system within the measurement manifold.

Sensitivity Structure

System sensitivity is derived from the information-geometric structure induced by the measurement model $p(m | d)$. It quantifies how strongly system outputs change under perturbations in difficulty states and is determined by the Fisher information geometry rather than empirical curve fitting.

Generative Measurement Structure

System behavior is represented as a conditional probability distribution over observable outcomes. Observed outputs are interpreted as samples from this generative measurement structure, separating deterministic structure from stochastic variation.

Information-Geometric Structure

NCAP-10 uses information geometry to quantify the local distinguishability of system responses across the difficulty manifold. This structure defines how rapidly measurement distributions diverge as difficulty states vary.

Normalization and Cross-System Comparison

All systems are evaluated within a normalized representation induced by the equivalence relation $\equiv_\varepsilon$. This ensures that comparisons are invariant to implementation details, parameterization, or internal representation.

Response Structure

System response is defined as the mapping from difficulty states to probability distributions over observables. Any empirical parameterizations are secondary approximations of this underlying probabilistic structure and do not define it.

Regulatory Structure

NCAP-10 represents internal stabilization mechanisms as transformations within the admissible transformation set T . These transformations describe how systems preserve behavioral consistency under perturbations in difficulty.

Memory Stability Structure

Memory behavior is represented as the temporal evolution of informational states on the measurement manifold. Stability is defined as the persistence of structure under this evolution, expressed relative to the NCAP-10 invariant geometry.

System Capability Functional

Overall system capability is defined as a derived functional on equivalence classes in $\mathcal{N}/\equiv_\varepsilon$. It jointly depends on performance structure, stability structure, and sensitivity structure, and is not a psychometric construct or scalar trait.

Invariance Principle

NCAP-10 is invariant under admissible transformations in T . This ensures that all evaluations depend only on equivalence-class structure in $\mathcal{N}/\equiv_\varepsilon$ and not on internal implementation or representation.

7 NCAP-10 Invariance Theorem

[NCAP-10 Invariance Theorem] Let $N, N' \in \mathcal{N}$ be cognitive objects such that $N \equiv_\varepsilon N'$ under the equivalence relation induced by the admissible transformation set T . Then all observable operators are constant on their equivalence classes:

$$\Phi([N]) = \Phi([N']), \quad \Psi([N]) = \Psi([N']), \quad \Omega([N]) = \Omega([N']).$$

[Sketch of Proof] Since $N \equiv_\varepsilon N'$, there exists an admissible transformation $F \in T$ such that $N' = F(N)$ and all NCAP-10 invariance axioms apply. By observable-class invariance,

$$\Phi([N]) = \Phi([F(N)]) = \Phi([N']),$$

and the same argument holds for Ψ and Ω . Thus, each observable operator is invariant on equivalence classes in $\mathcal{N}/\equiv_\varepsilon$, establishing architecture-independent evaluation.

8 Management version of the Categorical NCAP-10 Invariance Theorem

The categorical formulation of the NCAP-10 invariance principle provides a high-level, architecture-independent view of how cognitive systems are compared within the framework. Rather than focusing on internal implementation details, the categorical perspective emphasizes relationships, transformations, and behavioral equivalence across different system designs.

Cognitive Systems as Objects

In this formulation, each cognitive system is treated as an *object* in an enriched category. This abstraction allows NCAP-10 to represent systems with different architectures, internal representations, or computational substrates in a unified manner.

Admissible Transformations as Morphisms

Transformations between systems are represented as *morphisms*. These morphisms correspond to admissible, bounded-distortion mappings that preserve the essential behavioral and geometric properties required by NCAP-10. They capture the idea that two systems may differ internally yet remain functionally comparable.

Approximate Equivalence

NCAP-10 defines an ε -approximate equivalence relation based on the existence of such bounded-distortion morphisms. This equivalence does not require strict identity or perfect isomorphism; instead, it reflects the practical notion that two systems may be considered equivalent if their differences fall within an acceptable tolerance.

Observables as Functors

The quantities NCAP-10 computes—such as capability measures, difficulty embeddings, and invariant signals—are represented as *functors* from the cognitive category to a structured output category. These functors ensure that observable quantities depend only on the system’s behavior and not on its internal representation.

Factorization Through Equivalence Classes

The invariance theorem states that each observable functor factors through the quotient category formed by grouping systems into equivalence classes. This means that all NCAP-10 measurements are constant on these classes and therefore insensitive to representational differences that do not affect behavior.

Implications for Evaluation

This categorical structure guarantees that NCAP-10 provides a consistent and representation-agnostic evaluation of cognitive systems. Systems that behave similarly—up to allowable distortion—receive identical observable outputs, ensuring fairness and comparability across diverse architectures.

9 Categorical NCAP-10 Invariance Theorem

To express NCAP-10 invariance in categorical form, define the ε -enriched category $\mathbf{NCAP}_\varepsilon$ as follows:

- Objects are cognitive objects $N \in \mathcal{N}$.
- Morphisms are admissible ε -bounded distortion maps $F : N \rightarrow N'$ satisfying the NCAP-10 invariance axioms:

$$F^* g_I \sim_\varepsilon g_I, \quad p(m \mid d) = p(F(m) \mid F(d)).$$

- Each hom-set $\text{Hom}(N, N')$ is equipped with a distortion measure $\delta(F)$, making $\mathbf{NCAP}_\varepsilon$ an enriched category over $(R_{\geq 0}, \leq)$.

NCAP-10 equivalence $N \equiv_\varepsilon N'$ is defined as the existence of a morphism $F : N \rightarrow N'$ with $\delta(F) \leq \varepsilon$. This is *not* strict isomorphism but an ε -approximate equivalence relation.

Let $\mathbf{NCAP}_\varepsilon / \sim$ denote the quotient enriched category obtained by identifying objects related by $\equiv_\varepsilon$.

[Categorical NCAP-10 Invariance] Let

$$\Phi, \Psi, \Omega : \mathbf{NCAP}_\varepsilon \rightarrow \mathbf{Obs}$$

be observable functors into a structured codomain category $\mathbf{Obs}$ (e.g., metric spaces, manifolds, or ordered sets), each satisfying the NCAP-10 invariance axioms. Then each functor factors uniquely through the quotient enriched category $\mathbf{NCAP}_\varepsilon / \sim$:

$$\mathbf{NCAP}_\varepsilon[rr, " \Phi "] [dr, " \pi "'] \mathbf{ObsNCAP}_\varepsilon / \sim [ur, " \tilde{\Phi} "']$$

and similarly for Ψ and Ω .

Equivalently, for any $N \equiv_\varepsilon N'$,

$$\tilde{\Phi}([N]) = \tilde{\Phi}([N']), \quad \tilde{\Psi}([N]) = \tilde{\Psi}([N']), \quad \tilde{\Omega}([N]) = \tilde{\Omega}([N']).$$

[Sketch of Proof] Since $N \equiv_\varepsilon N'$, there exists an admissible morphism $F : N \rightarrow N'$ with $\delta(F) \leq \varepsilon$. By the invariance axioms, each observable functor sends F to an identity morphism in $\mathbf{Obs}$:

$$\Phi(F) = \text{id}_{\Phi(N)}, \quad \Psi(F) = \text{id}_{\Psi(N)}, \quad \Omega(F) = \text{id}_{\Omega(N)}.$$

Thus, each functor is constant on ε -equivalence classes and therefore factors uniquely through the quotient enriched category. This establishes invariance under all admissible bounded-distortion transformations.

Section Summary

The categorical invariance theorem formalizes the principle that NCAP-10 evaluates systems based on their functional behavior rather than their internal form. By expressing this principle through objects, morphisms, functors, and quotient categories, the framework ensures that all measurements are stable, comparable, and independent of implementation details.

10 Estimation Procedures

NCAP-10 estimation procedures specify how elements of the cognitive object space $\mathcal{N}$ and their induced geometric structures are recovered from empirical observations. All estimation is performed at the level of probability distributions and their associated information geometry, and all results are defined on equivalence classes $\mathcal{N}/\equiv_\varepsilon$ to ensure architecture-invariant recovery.

10.1 Estimating the Measurement Model

The conditional measurement distribution $p(m \mid d)$ is estimated using a parametric or non-parametric density model $\hat{p}(m \mid d)$ obtained by maximizing the expected log-likelihood:

$$\hat{p} = \arg \max_{p_\theta} E_{(m,d)}[\log p_\theta(m \mid d)].$$

This estimator defines the generative response structure of the system under varying difficulty states.

10.2 Estimating the Information Geometry

The Fisher information metric is estimated from the learned measurement model via

$$\hat{g}_I(d) = E_{m \sim \hat{p}(\cdot \mid d)}[\nabla_d \log \hat{p}(m \mid d) \nabla_d \log \hat{p}(m \mid d)^\top].$$

This induces the local distinguishability structure on the difficulty manifold and determines the system’s geometric sensitivity profile.

10.3 Estimating Sensitivity Structure

System sensitivity is defined as a functional of the estimated information geometry:

$$\hat{S}(d) = \mathcal{F}(\hat{g}_I(d)),$$

where $\mathcal{F}$ is a geometry-preserving functional summarizing local response variability. This avoids direct differentiation of empirical measurements and ensures stability under representation changes.

10.4 Estimating Response Structure

The response structure is defined as the conditional expectation under the estimated measurement model:

$$\mu(d) = E_{m \sim \hat{p}(\cdot|d)}[m].$$

Any parametric response curve (e.g., power-law models) is treated as an approximation to this expectation rather than a primary definitional object.

10.5 Estimating Normalized Representation

All systems are mapped into a normalized coordinate system induced by the equivalence relation $\equiv_\varepsilon$. This ensures invariance under admissible transformations:

$$\hat{p}(m | d) \sim \hat{p}(F(m) | F(d)), \quad F \in T.$$

Thus, estimation is performed on equivalence classes $[N] \in \mathcal{N}/\equiv_\varepsilon$ rather than on raw representations.

10.6 Estimating Memory Stability

Memory stability is defined on the temporal evolution of the conditional measurement model. Let $\hat{p}_t(m | d)$ denote the time-indexed distribution. Stability is estimated as

$$\hat{S}_{\text{mem}} = E \left[\exp \left(- \int_0^T \rho(t) dt \right) \right],$$

where $\rho(t)$ is the decay rate of the conditional information structure inferred from $\hat{p}_t$. This quantifies the persistence of informational states over time.

10.7 Estimating the Capability Functional

The system capability functional is defined on equivalence classes as a joint functional of the information geometry and measurement structure:

$$\hat{\Theta}(N) = \mathcal{G}(\hat{g}_I, \hat{p}),$$

where $\mathcal{G}$ is invariant under all admissible transformations $F \in T$. This functional is defined on $\mathcal{N}/$

11 Economic Functor Layer

NCAP-10 admits an economic interpretation formulated as a structure-preserving functor from the invariant quotient space $\mathcal{N}/\equiv_\varepsilon$ to a structured economic category. This ensures that all economic quantities depend only on equivalence classes and remain invariant under admissible transformations.

Formally, define the economic functor

$$\mathcal{E} : \mathcal{N}/\equiv_\varepsilon \longrightarrow \mathbf{Econ},$$

where $\mathbf{Econ}$ is a category whose objects are structured economic states and whose morphisms represent monotone, resource-compatible transformations preserving ordering of value, risk monotonicity, and cost scaling.

11.1 Economic Value of Capability

For an equivalence class $[N]$, the economic value is defined as a class-functional:

$$V([N]) = E_{m \sim P_{[N]}}[R(m)] - C([N]),$$

where:

- $P_{[N]}$ is the induced measure class associated with $[N]$, i.e., the set of all measurement distributions consistent with representatives of the equivalence class,
- $R(m)$ is a reward functional on observable outcomes,
- $C([N])$ is a cost functional defined on equivalence classes.

This formulation removes dependence on any representative d and ensures full invariance under $\mathcal{N}/\equiv_\varepsilon$.

11.2 Marginal Value Structure (Invariant Form)

NCAP-10 defines marginal economic variation through counterfactual movement in the quotient space. Let $[N][N']$ denote an admissible perturbation in $\mathcal{N}/\equiv_\varepsilon$. The marginal value is:

$$\Delta V = V([N']) - V([N]).$$

This defines economic sensitivity without requiring non-structural derivatives of observables.

11.3 Risk as Information-Geometric Volume

Operational risk is defined as an isomorphism-invariant scalar functional of the Fisher geometry:

$$\text{Risk}([N]) = \mathcal{R}(g_I),$$

where $\mathcal{R}$ is invariant under coordinate transformations of the underlying manifold.

A canonical invariant choice is the Riemannian volume form:

$$\text{Risk}([N]) = \log \int_X \sqrt{\det(g_I)} dx,$$

which depends only on intrinsic geometric dispersion and is independent of parametrization.

11.4 Return on Resources

Let $C([N])$ denote a resource expenditure functional (compute, energy, or time). The return on resources is defined as:

$$\text{RoR}([N]) = \frac{V([N])}{C([N])}.$$

This quantity is defined entirely on equivalence classes and is invariant under admissible transformations.

11.5 Control Cost as Geometric Deformation

Control actions correspond to admissible morphisms $F : [N] \rightarrow [N']$ in the quotient category. The cost of control is defined as geometric distortion of the Fisher structure:

$$J(F) = d_{\mathcal{G}}(g_I, F^* g_I),$$

where $d_{\mathcal{G}}$ is a metric on the space of Riemannian metrics.

The optimal control action is:

$$F^* = \arg \min_{F \in T} J(F).$$

This defines control as a geometry-preserving optimization problem over admissible transformations.

11.6 Memory Value Retention

Let $[N_t]$ denote a time-indexed family of equivalence classes and let $\rho(t)$ denote the intrinsic decay rate of the conditional information structure induced by $\hat{p}_t$. Memory value retention is defined as:

$$V_{\text{mem}}([N]) = E \left[\exp \left(- \int_0^T \rho(t) dt \right) \right].$$

This formulation depends only on the evolution of the information-geometric structure and is independent of representation.

11.7 Functorial Invariance Principle

The economic functor is well-defined on the NCAP-10 quotient category:

$$\mathcal{E} : \mathcal{N} / \equiv_{\varepsilon} \rightarrow \mathbf{Econ},$$

and satisfies naturality:

$$\mathcal{E} \circ F = \mathcal{E}, \quad \forall F \in T.$$

Equivalently, for all equivalence classes $[N]$ and $[N']$ such that $[N] = [N']$,

$$\mathcal{E}([N]) = \mathcal{E}([N']).$$

Thus, economic evaluation is a functorial invariant of NCAP-10 and is fully independent of representation, architecture, and coordinate choice.

12 Management Summary of the Economic Functor Layer

The Economic Functor Layer provides a high-level interpretation of NCAP-10's geometric framework in economic terms. Rather than treating economics as an external add-on, this layer defines a structure-preserving mapping from NCAP-10's invariant cognitive object space to an abstract economic space. This ensures that all economic quantities remain consistent, comparable, and independent of system architecture.

Economics as a Functor

NCAP-10 represents each cognitive system as an equivalence class $[N]$ that captures its behavioral and geometric properties. The economic layer defines a functor

$$\mathcal{E} : \mathcal{N} / \equiv_{\varepsilon} \rightarrow \mathbf{Econ},$$

which assigns to each equivalence class an economic state containing value, cost, risk, and efficiency measures. Because the mapping is functorial, all economic outputs remain invariant under admissible transformations.

Economic Value of Capability

The value of a system is defined as the expected reward from its observable behavior minus the cost of operating at a given difficulty level. Both reward and cost are defined on equivalence classes, ensuring that economic value depends only on behavior, not on internal representation.

Marginal Value Through Counterfactual Variation

NCAP-10 does not assume differentiability of measurements. Instead, marginal economic effects are defined through counterfactual comparisons between nearby equivalence classes. This provides a robust and representation-free notion of economic sensitivity.

Risk from Information Geometry

Operational risk is derived from the Fisher information geometry. An invariant functional of the metric—such as the log-determinant—quantifies how sensitive a system is to perturbations. Higher geometric dispersion corresponds to higher operational risk.

Return on Resources

Return on resources is defined as the ratio of economic value to resource expenditure. Because both terms are defined on equivalence classes, this measure captures the intrinsic efficiency of a system independent of architecture or implementation details.

Control Cost as Geometric Distortion

Control actions correspond to admissible transformations between equivalence classes. The cost of control is defined as the geometric distortion induced by such a transformation. Optimal control minimizes this distortion, ensuring interventions are efficient and minimally disruptive.

Memory Value Retention

Memory value is defined through the time evolution of the system’s information geometry. An exponential decay functional quantifies how much value is retained over time, linking memory stability directly to economic depreciation.

Functorial Invariance

All economic quantities satisfy the invariance principle:

$$\mathcal{E}([N]) = \mathcal{E}([N']) \quad \text{whenever} \quad [N] = [N'].$$

This ensures that economic evaluation is fully architecture-independent and consistent with the core NCAP-10 invariance structure.

Summary

The Economic Functor Layer provides a rigorous and invariant bridge between NCAP-10’s geometric framework and economic interpretation. By defining

value, risk, cost, and efficiency as functorial outputs, the framework ensures that economic assessments remain stable, comparable, and grounded in the intrinsic behavior of cognitive systems.

13 Discussion

NCAP-10 establishes a unified, coordinate-free framework for evaluating machine cognition by integrating geometric, probabilistic, and economic structures into a single invariant system. All quantities are defined on the quotient space $\mathcal{N}/\equiv_\varepsilon$, ensuring that evaluation is independent of architecture, representation, and parameterization. Several fundamental consequences follow.

(1) Invariance Implies Comparability

NCAP-10 is defined on equivalence classes $[N] \in \mathcal{N}/\equiv_\varepsilon$, where admissible transformations act as structure-preserving morphisms:

$$F \in T \quad \Rightarrow \quad F(N) \sim_\varepsilon N.$$

As a result, all systems—regardless of internal implementation—are embedded into a single statistical-geometric space. This removes representation-induced bias and enables direct comparisons across heterogeneous architectures.

(2) Geometry Implies Risk Structure

The Fisher information metric g_I defines an intrinsic geometric structure on the difficulty manifold. Operational risk is an isomorphism-invariant functional of this geometry:

$$\text{Risk}([N]) \equiv \mathcal{R}(g_I).$$

A canonical invariant realization is the Riemannian volume functional:

$$\text{Risk}([N]) = \log \int_X \sqrt{\det(g_I)} dx.$$

Thus, risk is not externally imposed but emerges as intrinsic geometric dispersion of the information manifold.

(3) Scaling Laws Imply Structural Economic Constraints

Empirical scaling laws are absorbed into the invariant response structure $m^\dagger(d)$ and interpreted as constraints on the geometry of achievable performance. Power-law behavior induces fundamental diminishing returns in the induced economic functional space, implying that increases in resource allocation yield sublinear gains in invariant capability.

Thus, scaling behavior is not merely empirical but a consequence of the underlying geometric structure of $\mathcal{N}/\equiv_\varepsilon$.

(4) Control as Geometric Optimization

Control is formulated as optimization over admissible morphisms in the quotient category. A control action corresponds to a transformation of geometric structure:

$$F : [N] \rightarrow [N'], \quad F \in T.$$

The associated control cost is the induced deformation of the Fisher geometry:

$$J(F) = d_{\mathcal{G}}(g_I, F(g_I)),$$

where $d_{\mathcal{G}}$ is a metric on the space of Riemannian structures. The optimal control action is therefore:

$$F^* = \arg \min_{F \in T} d_{\mathcal{G}}(g_I, F(g_I)).$$

Control is thus a geometric projection problem onto admissible invariant transformations.

(5) Memory as Geometric Decay of Information Structure

Memory is defined on time-indexed equivalence classes $[N_t]$, where the evolution of the induced information geometry determines retention. Let $\rho(t)$ denote the intrinsic decay rate of the conditional information structure induced by $p_t(m | d)$. Memory value retention is defined as:

$$V_{\text{mem}}([N]) = E \left[\exp \left(- \int_0^T \rho(t) dt \right) \right].$$

This formulation identifies memory as a survival functional over trajectories in $\mathcal{N}/\equiv_{\varepsilon}$, fully invariant under admissible transformations.

Section Summary

NCAP-10 reduces machine cognition evaluation to a coordinate-free geometric-economic theory on the quotient space $\mathcal{N}/\equiv_{\varepsilon}$. Invariance guarantees comparability across architectures, Fisher geometry defines intrinsic risk, scaling laws impose structural economic constraints, control becomes geometric optimization over admissible morphisms, and memory emerges as a decay functional on evolving information geometry.

The resulting framework is a fully invariant, architecture-independent theory of cognitive measurement grounded in geometry, probability, and category-theoretic structure.

14 Conclusion

NCAP-10 provides a closed, invariant, and geometry-based measurement theory for machine cognition. By grounding evaluation in the quotient space $\mathcal{N}/\equiv_\varepsilon$, the framework ensures that all assessments are independent of architecture, representation, or implementation details. This yields a unified basis for cross-system comparability, safety analysis, and economic interpretation.

NCAP-10 demonstrates that cognition can be decomposed into four invariant functional components defined on equivalence classes $[N] \in \mathcal{N}/\equiv_\varepsilon$:

- **Geometry:** the information-geometric structure induced by g_I on X , determining sensitivity and distinguishability of system responses.
- **Scaling:** asymptotic response structure governing performance under increasing difficulty, constraining achievable capability growth.
- **Control:** admissible transformations $F \in T$ acting as structure-preserving morphisms over the quotient space.
- **Memory:** time-indexed evolution of the induced information structure, defining stability and retention through geometric decay.

Formally, cognition is represented at the level of invariant structure as:

$$\mathcal{C}([N]) \equiv (\mathcal{G}[N], \mathcal{S}[N], \mathcal{U}[N], \mathcal{M}[N]),$$

where each component is a well-defined functional on the equivalence class $[N]$, invariant under all admissible transformations.

All economic value, safety properties, and capability measures are therefore determined by invariant functionals on $\mathcal{N}/\equiv_\varepsilon$, rather than by representational details of individual systems. In particular:

- evaluation is invariant and architecture-neutral,
- optimization is a geometric variational problem over admissible morphisms,
- governance corresponds to measurable transformations in the quotient category.

The framework is closed under invariance in the sense that all observables, economic quantities, and control actions factor through the quotient structure:

$$\mathcal{N} \longrightarrow \mathcal{N}/\equiv_\varepsilon \longrightarrow \mathbf{Econ}.$$

Any further reduction of structure would require relaxing the NCAP-10 invariance axioms or discarding the underlying information-geometric representation. Within these constraints, NCAP-10 is structurally minimal: all admissible evaluations factor through invariant geometry, and no representation-dependent degrees of freedom remain.

In summary, NCAP-10 establishes a principled, invariant, and economically interpretable foundation for evaluating machine cognition, providing a unified framework suitable for scientific, commercial, and regulatory use grounded in quotient geometry, information structure, and functorial consistency.

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ABSTRACT

: NCAP-10 Invariance Theorem Full Proof

[NCAP-10 Invariance Theorem] Let $N, N' \in \mathcal{N}$ be cognitive objects such that $N \equiv_\varepsilon N'$ under the equivalence relation induced by the admissible transformation set T . Assume that observable operators are defined on equivalence classes

$$\Phi, \Psi, \Omega : \mathcal{N} / \equiv_\varepsilon \rightarrow \mathbf{Obs},$$

and are well-defined in the sense that their values do not depend on the choice of representative. Then

$$\Phi([N]) = \Phi([N']), \quad \Psi([N]) = \Psi([N']), \quad \Omega([N]) = \Omega([N']).$$

By definition of NCAP-10 equivalence, $N \equiv_\varepsilon N'$ means that there exists an admissible transformation $F \in T$ such that

$$N' = F(N),$$

and F satisfies the NCAP-10 invariance constraints at the level of geometry and measurement. Concretely, F preserves the information structure up to bounded distortion:

$$F^* g_I \sim_\varepsilon g_I,$$

and preserves the conditional measurement behavior up to the same distortion regime, i.e. within the tolerance encoded by $\equiv_\varepsilon$. The precise form of this preservation (exact or approximate) is absorbed into the definition of the equivalence relation and the admissible set T .

The quotient space $\mathcal{N} / \equiv_\varepsilon$ is constructed by identifying all cognitive objects related by such admissible transformations. Thus $N \equiv_\varepsilon N'$ implies

$$[N] = [N'] \quad \text{in} \quad \mathcal{N} / \equiv_\varepsilon.$$

By assumption, each observable operator

$$\Phi, \Psi, \Omega : \mathcal{N} / \equiv_\varepsilon \rightarrow \mathbf{Obs}$$

is defined on equivalence classes and is independent of the choice of representative. Therefore,

$$\Phi([N]) = \Phi([N']), \quad \Psi([N]) = \Psi([N']), \quad \Omega([N]) = \Omega([N']).$$

In other words, once observables are defined on the quotient space $\mathcal{N} / \equiv_\varepsilon$, their invariance under admissible transformations is immediate: they are constant on equivalence classes by construction. The theorem thus formalizes the structural fact that NCAP-10 observables factor through the quotient induced by $\equiv_\varepsilon$ and depend only on behavioral equivalence, not on representational form.